B.Tech.(IT) Major Project–I (IT449) Mid-Semester Evaluation

R.E.A.P : Realtime Edge Agentic Physio-Buddy

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Table of Contents

- Introduction
- Market Research
- Literature Survey
- Outcome of the Literature Survey
- Problem Statement & Objectives
- Software Requirement Specification (Functional & Non-Functional)
- Timeline
- Methodology
- References

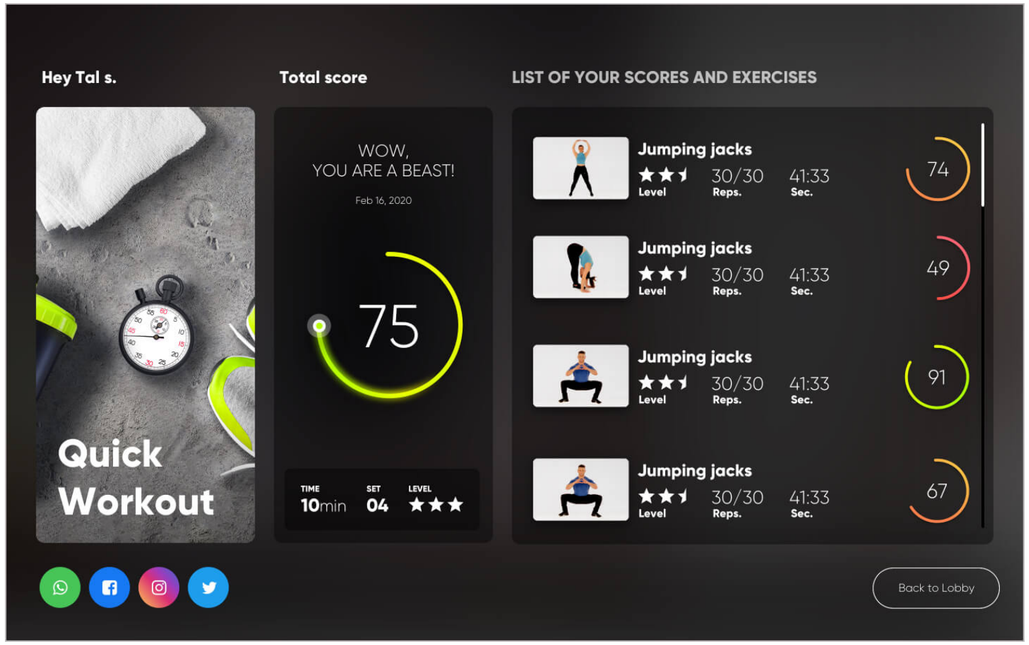
Introduction :

Silent Degradation: Posture and technique break down long before physical fatigue is felt.
High Injury Risk: Severe joint and tendon strain occurs when form degrades mid-set, particularly for older adults.
Blind Training: Self-correction in real-time is nearly impossible without external visual oversight or a coach.
The Need: Trainees require instant, honest auditory feedback for every single rep to prevent devastating injuries.



Market Research :

QuickPose: Relies on 100% on-device MediaPipe SDK, but uses generic heuristics unsuitable for targeted mobility.
Kemtai: A 111-point clinical platform validated against labs, but lacks focus on heavy compound mechanics.
Metric VBT: Tracks barbell velocity at 60 FPS, but completely ignores lifter biomechanics and joint stress.



Literature Survey

“N-euro Predictor: A Neural Network Approach for Smoothing and Predicting Motion Trajectory”

Key Points

The Jitter-Lag Tradeoff: Traditional motion smoothing algorithms (such as the 1 Euro filter) inherently create a compromise; reducing spatial jitter (noise) requires a longer sampling window, which introduces unacceptable rendering lag (Shao et al., 2023).

Predictive Neural Modeling: N-euro replaces reactive filtering with a lightweight, residual-learning neural network that predicts near-future joint positions based on short-term historical dynamics (Shao et al., 2023).

Application to Your Project: By rendering the predicted trajectory rather than the delayed raw sensor data, N-euro achieves true sub-second latency, which is critical for delivering the mid-rep auditory cues your system relies on to prevent injuries.

Literature Survey

“Spatio-Temporal Graph Convolution for Skeleton Based Action Recognition”

Key Points

Beyond Isolated Joints: Rather than tracking individual joints with hardcoded thresholds, ST-GCN models the human skeleton as a continuous dynamic graph. It connects spatial edges (anatomical connectivity) and temporal edges (the same joints across consecutive frames).

Holistic Action Recognition: This architecture allows the system to automatically capture patterns embedded in the spatial configuration of the joints as well as their temporal dynamics without relying on hand-crafted rules.

Application to Your Project: ST-GCN enables your system to instantly recognize systemic kinetic chain breakdowns (e.g., how a slight shift in the ankle impacts the knee and hip simultaneously) rather than waiting for a single isolated joint to cross a danger threshold.

Outcome of Literature Survey

Elimination of the Jitter–Lag Tradeoff: Traditional heuristic filters (like 1 Euro) introduce latency to smooth noise, whereas predictive neural filtering (N-euro) enables sub-second pose estimation needed for synchronous mid-rep auditory feedback.

Shift to Kinetic Chain Modeling: Spatial-Temporal Graph Convolutional Networks (ST-GCN) effectively replace rigid single-joint thresholds by analyzing full-body spatial and temporal joint relationships to detect holistic form breakdowns.

Necessity of Edge AI and Audio-First Interfaces: Relying on cloud processing introduces latency risks, while screen-based UIs compromise cervical spine alignment, establishing the requirement for fully local edge inference paired with multi-modal audio cues.

Problem Statement

- Older adults face a high risk of injury during mobility exercises because biomechanical form often degrades long before the individual consciously feels fatigue.
- To address this, the Real-Time Kinematic Form Corrector utilizes localized Edge AI and hybrid pose estimation to deliver sub-second, multi-modal auditory feedback.
- By proactively monitoring velocity-loss and kinetic chain deviations, our system provides instant, eyes-free mid-rep corrections, empowering users to self-correct or halt sets before critical mechanical failure occurs.



Main Objectives

Objective 1 (Instant Feedback): Deliver sub-second, multi-modal auditory corrections mid-rep.

Objective 2 (Injury Prevention): Proactively monitor velocity-loss and pose deviation to halt sets prior to mechanical failure.

Objective 3 (Enhanced Accessibility): Provide professional-grade kinematic tracking accessible via smartphone for older adults at home.

Software Requirement Specification (Functional)

Kinematic Tracking: Track multi-point joint angles and exercise trajectory continuously.
Deviation Analytics: Calculate mechanical deviations against clinically validated pose baselines.
Real-Time Audio: Deliver real-time text-to-speech (TTS) and audio cues instructing form correction.
Fatigue Profiling: Execute predictive fatigue profiling to proactively warn the user.

Software Requirement Specification (Non-Functional)

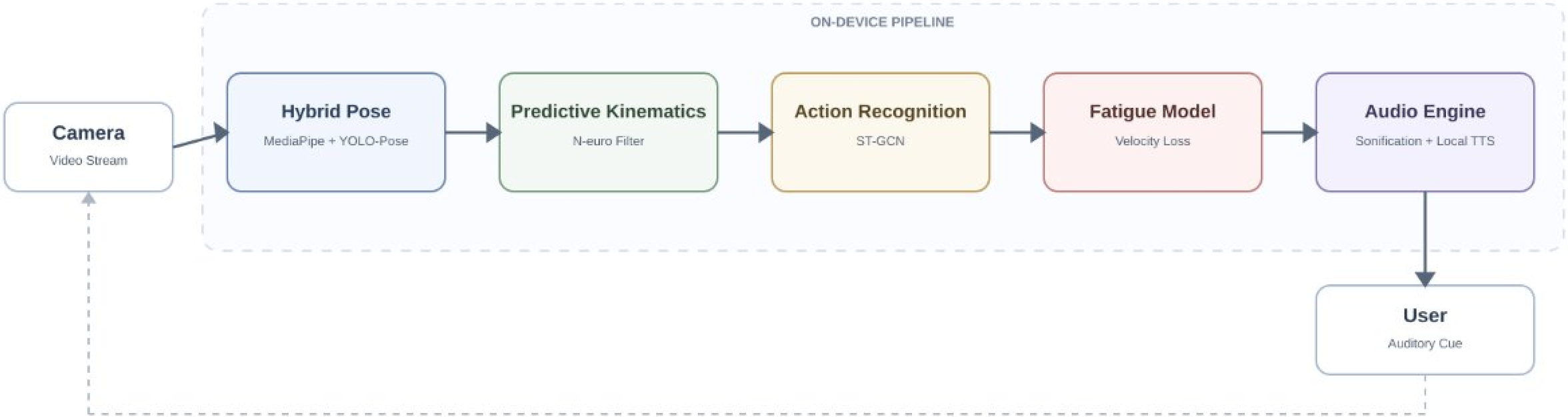
K Latency Limits: Maintain strict sub-second visual-to-audio processing pipelines.
Total Localization: Rely entirely on localized Edge AI with zero cloud dependency during lifts.
Audio-First Usability: Audio-first UI ensuring the user maintains cervical spine neutrality without looking at screens.
System Modularity: Built on a plug-in Python framework for adding new exercise profiles.

Methodology Used In this Proejct Work

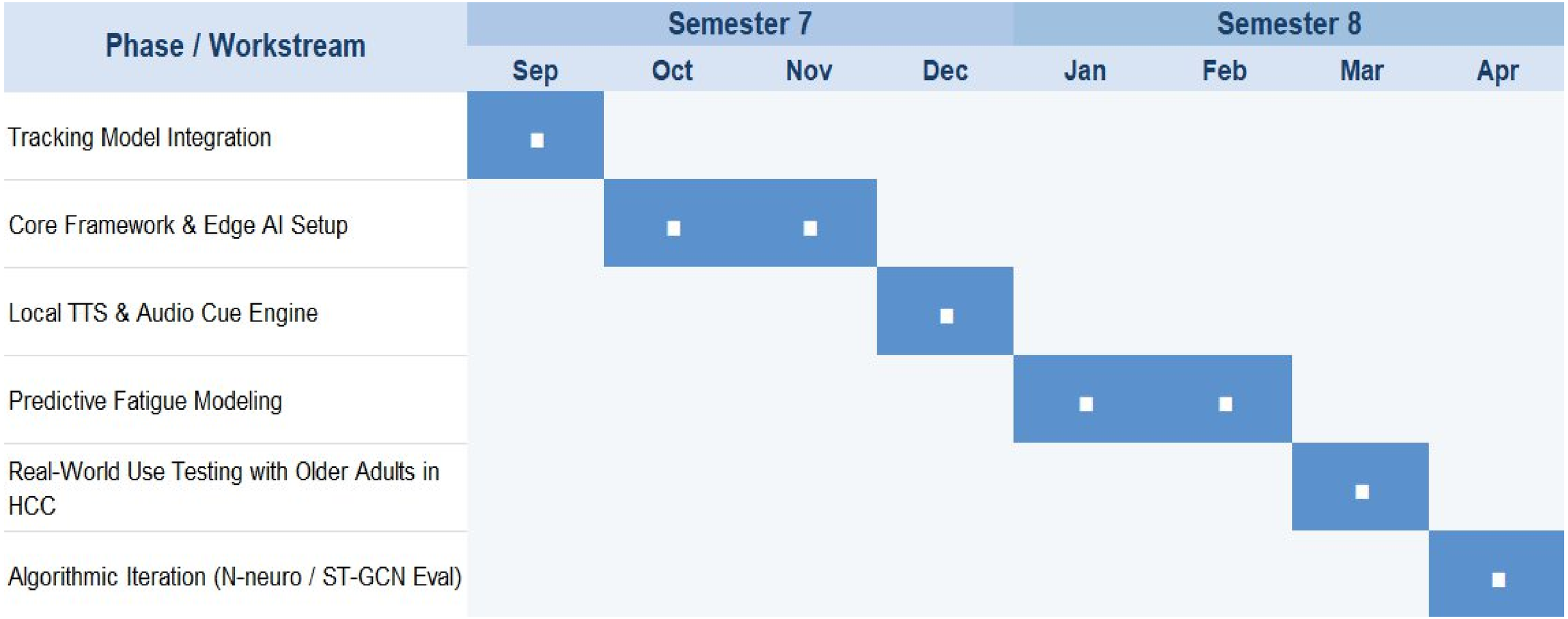
- **Kinematic Pipeline:** Fusing MediaPipe and YOLO-Pose for robust, real-time kinematic tracking of joint angles under various conditions.
- **Python Framework:** A modular application core managing continuous trajectory calculation and supporting "plug-in" exercise packs.
- **Local TTS Engine:** Generates sub-second auditory cues entirely on-device, bypassing network lag for mid-rep corrections.



Architecture



Timeline of the Project



References

1. Li, C., Cui, Z., Zheng, W., Xu, C., & Yang, J. Spatio-Temporal Graph Convolution for Skeleton Based Action Recognition. *Proceedings of the AAAI Conference on Artificial Intelligence*, *32*. https://doi.org/10.1609/aaai.v32i1.11776 Cited by: 7614

2. Shao, Q., Wang, J., Zhou, B., Tran, V. A., Krishnan, G., & Nayar, S. (2023). N-euro Predictor. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, *7*, 1-25. https://doi.org/10.1145/3610884 Cited by: 8